Beyond point-generated product–join constructions for projection-body volume

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Abstract

We give an explicit rational eighteen-dimensional seed whose iterated products and joins have projection-volume root limit strictly between 2.9280 and 2.9283. In combination with the prior bound $\Gamma_{\mathcal{C}} \le e^{1.049} < 2.855$ for the class generated from a point by products, joins, and affine equivalences, this proves a strict asymptotic separation between that entire class and unrestricted convex polytopes. Explicit finite product–join recipes and a fixed-block extension exceed every member of $\mathcal{C}_n$ for every $n \ge 9$. The principal seed is a codimension-four section of the standard simplex specified by twenty-three integer four-vectors. A simpler symmetric fifteen-dimensional section gives a limit above 2.890, a twelve-dimensional two-upper-facet seed gives a limit above 2.861, and a simpler eight-dimensional seed gives a limit above 2.8571. The section method, product–join calculus, recursive scheme, and comparison upper bound are attributed to prior work. The contribution is the specified rational seeds, their stronger certified amplified rates, and the class separation. Exact geometry, Laplace-transform identities, and recurrence certificates accompany the proof; global optimality and bibliographic priority are not asserted.

1 The separation problem and main result

For a full-dimensional convex body $K \subset \mathbb{R}^n$, use the normalization

$$h_{\Pi K}(u) = \mathrm{vol}_{n-1}(K \mid u^{\perp}), \qquad R_n(K) = \frac{\mathrm{vol}_n(\Pi K)}{\mathrm{vol}_n(K)^{n-1}}.$$

Here u is a unit vector and $K \mid u^{\perp}$ is its orthogonal projection. Write $g(n) = n^n/n!$ and $c_n = (n+1)g(n) = R_n(T_n)$, where T_n is an n-simplex. The ratio is invariant under invertible affine maps. For standard background on support functions, area measures, and affine convex geometry, see Schneider [17]. We omit the dimension subscript when it is unambiguous. The standard product and join are taken in independent coordinate spaces:

$$A * B = \mathrm{conv}\big((A,0,0) \cup (0,B,1)\big).$$

Let $\mathcal{C}$ consist of polytopes obtained from a zero-dimensional point by finitely many Cartesian products, joins, and invertible affine maps on their affine hulls. Rank-dropping affine maps are outside this definition. Set $\mathcal{C}_n = \{K \in \mathcal{C} : \dim K = n\}$ and

$$\Gamma_{\mathcal{C}} = \limsup_{n\to\infty} \sup_{K\in\mathcal{C}_n} R_n(K)^{1/n}.$$

The limit in fact exists by the spectral reduction in [12], but the limsup definition suffices below. Let U_n be the supremum of $R_n(K)^{1/n}$ over all full-dimensional convex bodies $K \subset \mathbb{R}^n$.

The unrestricted maximization problem has a longer history. Brannen [1] formulated the simplex maximum conjecture and disproved a different proposed upper bound for centrally symmetric bodies. Lutwak, Yang, and Zhang [7] developed affine-invariant functionals for polytopes and derived projection-volume estimates. Saroglou [15] established sharp three-dimensional maxima within the classes of zonoids, cones, and double cones. These results concern the same volume ratio, but do not supply the whole-class comparison used here.

Feng, Hu, Liu, and Xu [4] already disprove the simplex upper bound in every dimension at least nine using polytopes with at most $n+2$ facets. OpenAI's product example [14] provides a particularly simple later counterexample. Subsequent work in the mxym repository gives the exact product–join calculus and a recursive construction with root limit between 2.8534 and 2.8535 [13, Section 9]. Its mixed Bellman bound [10] controls every finite expression in $\mathcal{C}$, not just that particular recursion. The question addressed here is whether the point-generated closure itself can attain the best unrestricted exponential rate.

Theorem 1. *There is an explicitly specified sequence of rational polytopes $P_n \subset \mathbb{R}^n$ such that*

$$\lim_{n\to\infty} R_n(P_n)^{1/n} = \beta, \qquad 2.9280 < \beta < 2.9283.$$

Consequently, using the prior upper theorem [10],

$$\Gamma_{\mathcal{C}} < 2.855 < \beta \le \liminf_{n\to\infty} U_n.$$

There is also an explicit sequence E_n, given by finite product–join recipes and a fixed-block extension, satisfying

$$R_n(E_n) > \sup_{K\in\mathcal{C}_n} R_n(K) \qquad \text{for every } n \ge 9.$$

The sequence P_n can be chosen to satisfy the same strict inequality for every $n \ge 9$. No minimal-dimension claim is made.

Thus the advantage concerns an entire construction class. For example, the asymptotic ratio of $R_n(P_n)$ to its class supremum has lower root at least $2.928/2.855 > 1.0255$. This is an exponential separation; it is not a claim of optimality among all convex bodies. For context, Zhang's reverse polar projection inequality [18] and the Blaschke–Santaló inequality [9] give

$$R_n(K) \le \frac{\kappa_n^2 n^n}{\binom{2n}{n}}, \qquad \limsup_{n\to\infty} U_n \le \frac{\pi e}{2},$$

where κ_n is the volume of the Euclidean unit ball; see [16, Introduction, equation (3)] for the combined estimate. The limiting constant follows from Stirling's formula. A substantial unrestricted gap therefore remains.

2 Facet invariants and inherited operation rules

For a polytope K of dimension d and volume V, let v_i be its outward facet area-normal vectors and let ζ_i be the facet area times its support number. Equivalently, $\langle v_i, x\rangle = \zeta_i$ on facet i. Define the determinant sums

$$\mathcal{P}(K) = \sum_{|J|=d} |\det(v_j : j \in J)|, \qquad \mathcal{S}(K) = \sum_{|J|=d+1} |\det((v_j, \zeta_j) : j \in J)|.$$

Cauchy's projection formula gives $\mathcal{P}(K) = \mathrm{vol}_d(\Pi K)$: the projection body is a translate of the zonotope $\sum_i [0, v_i]$. The determinant expression for zonotope volume and its application to projection bodies are classical; see [7, Section 1] and [16, Section 3]. The lifted invariant of [13, Section 2] is

$$a(K) = \frac{\mathcal{S}(K)}{dV\mathcal{P}(K)}, \qquad H(K) = \frac{1}{a(K)}, \qquad Q(K) = \frac{a(K)R_d(K)}{g(d)}. \tag{1}$$

It is affine invariant and satisfies $1/(d+1) \le a(K) \le 1$. We use different symbols for the geometric tilt parameters below and the invariant a.

For positive-dimensional factors A, B of dimensions r, s, the exact identities in [13, Sections 3–5] are

$$R(A \times B) = R(A)R(B), \qquad a(A \times B) = \frac{ra(A) + sa(B)}{r+s}, \tag{2}$$

$$R(A * B) = \frac{g(r+s+1)}{g(r)g(s)} R(A)R(B)(a(A) + a(B)), \qquad a(A * B) = \frac{a(A)a(B)}{a(A) + a(B)}. \tag{3}$$

These rules apply to the nonsimplex seeds as well as to members of $\mathcal{C}$. They, and the balanced recursion used below, are imported prior results rather than new operation formulas.

3 An explicit rational seed

Let $n = 2m$ with $m \ge 1$, choose $0 < u_1 < \cdots < u_m < 1$, and put

$$b = (-u_m, \ldots, -u_1, u_1, \ldots, u_m), \qquad K(u) = \{x \ge 0 : \ \mathbf{1} \cdot x + b \cdot x \le 1, \ \mathbf{1} \cdot x - b \cdot x \le 1\}.$$

For rational u, this is a rational realization of the central two-facet section family underlying [4, Section 4]. We do not claim the family itself. The symmetry exchanging paired coordinates interchanges the two upper facets; it is not central symmetry of $K(u)$.

Proposition 2. *The polytope $K(u)$ has $n + 1 + m^2$ vertices and $n + 2$ facets. Put*

$$I(u) = \sum_{i=1}^{m} \frac{u_i^{2m-2}}{(1+u_i)\prod_{j \ne i}(u_i^2 - u_j^2)}, \qquad p(u) = n + 1 + \sum_{j=1}^{m} (2j-1)u_j, \qquad s(u) = 1 + \sum_{j=1}^{m} u_j. \tag{4}$$

Then

$$n!\,\mathrm{vol}_n(K(u)) = I(u), \qquad R_n(K(u)) = g(n)I(u)p(u), \qquad a(K(u)) = \frac{s(u)}{p(u)}. \tag{5}$$

In particular $Q(K(u)) = I(u)s(u)$.

Proof. The vertices are the origin, the n points $e_i/(1+|b_i|)$, and, for each pair $b_i = -v < 0 < b_j = w$,

$$\frac{w}{v+w} e_i + \frac{v}{v+w} e_j.$$

An active-set argument proves completeness: at a nonzero vertex one or both upper facets must be active, leaving at most one or two positive coordinates, respectively. In the latter case the signs must be opposite. Each coordinate hyperplane contains the origin and $n-1$ independent axis vertices. Each upper hyperplane contains a point with all coordinates positive and $b \cdot x$ of the required strict sign. Thus every listed inequality defines a facet.

On $b \cdot x \ge 0$, the upper constraint with $+b$ determines feasibility; the other upper constraint is redundant there. Change variables $y_i = (1 + b_i)x_i$ and use the standard simplex with slack

coordinate $y_0 = 1 - \sum_{i=1}^{n} y_i$. Its truncating coefficients are $t_0 = 0$ and $t_i = b_i/(1+b_i)$. The Jacobian is $\prod_i (1+b_i)^{-1}$ and $t_i - t_j = (b_i - b_j)/((1+b_i)(1+b_j))$ for $i, j \geq 1$. The simplex truncation identity in Appendix A, with these factors cancelled, gives

$$n!\,\mathrm{vol}_n(K(u) \cap \{b \cdot x \geq 0\}) = \sum_{b_i > 0} \frac{b_i^{n-1}}{(1+b_i)\prod_{j \neq i}(b_i - b_j)}.$$

The two halves have equal volume. Pairing $b_j = \pm u_j$ in the last denominator gives (4).

Write $V = \mathrm{vol}_n(K(u))$. Each upper facet vector is a positive multiple of $\mathbf{1} \pm b$. Reflection makes the two multiples equal. The divergence identity $\sum_i \zeta_i = nV$, with zero support on coordinate facets, makes each multiple $nV/2$. Balance of area-normal vectors then gives $-nVe_i$ on coordinate facet i. After division by nV, the horizontal generators and their lifted support coordinates are

$$(-e_i, 0), \qquad \left(\frac{\mathbf{1}+b}{2}, \frac{1}{2}\right), \quad \left(\frac{\mathbf{1}-b}{2}, \frac{1}{2}\right).$$

The horizontal n-minor sum is

$$1 + n + \frac{1}{2}\sum_{i<j} |b_i - b_j| = n + 1 + \sum_{j=1}^{m} (2j-1)u_j = p(u).$$

The lifted $(n+1)$-minor sum is $1 + \frac{1}{2}\sum_i |b_i| = s(u)$. Restoring the factors $(nV)^n$ and $(nV)^{n+1}$ in the two sums and using (1) proves (5). □

Our two-upper-facet seed is the following body with 49 vertices and 14 facets:

$$K_0 = K\left(\frac{17}{100}, \frac{24}{100}, \frac{32}{100}, \frac{41}{100}, \frac{53}{100}, \frac{69}{100}\right). \tag{6}$$

In integer form its upper inequalities are

$$\sum_i (100 \pm B_i)x_i \leq 100, \qquad B = (-69, -53, -41, -32, -24, -17, 17, 24, 32, 41, 53, 69).$$

Direct rational evaluation, independently checked from its geometry, gives

$$I_0 = \frac{1031929453906486710787861599001}{2386952337826268810834539955652}, \tag{7}$$

$$p_0 = \frac{768}{25}, \qquad s_0 = \frac{84}{25}, \qquad a_0 = \frac{7}{64}, \qquad R_0 = g(12)I_0\frac{768}{25}. \tag{8}$$

For comparison, the simpler seed $K(1/10, 2/10, 3/10, 4/10)$ in dimension eight has 25 vertices and

$$I_0 = \frac{231953}{378378}, \qquad p_0 = 14, \qquad s_0 = 2, \qquad a_0 = \frac{1}{7}.$$

Its initial ratio $R_8/c_8 = 231953/243243$ is below one. Nevertheless, the same recursive amplification already exceeds the asymptotic ceiling for $\mathcal{C}$. A seed's immediate projection-volume root therefore does not by itself describe its value under repeated operation rules.

4 A symmetric polygon-section seed

We enlarge the seed family to sections with three more facets than their dimension. Let $A(x, y) = (-y, x + y)$ and let $b_1, \dots, b_{18}$ be the columns

$$A^j(10, 0), \quad A^j(7, 4), \quad A^j(4, 7), \qquad 0 \le j < 6.$$

Write $B = (b_1, \dots, b_{18})$ and define the rational polytope

$$S_0 = \left\{ x \in \mathbb{R}^{18} : x_i \ge 0, \quad \sum_i x_i = 1, \quad Bx = 0 \right\}. \tag{9}$$

Its affine hull has dimension fifteen. Choosing fifteen free coordinates gives a rational affine realization in $\mathbb{R}^{15}$. Every coordinate inequality defines a facet: deleting any column leaves two independent antipodal pairs, so zero remains in the interior of the convex hull of the remaining columns. The family of simplex sections and the use of complementary minors follow the approach of [4, Section 4]; polygon configurations themselves are not claimed as new.

Proposition 3. *The seed S_0 has eighteen facets and exact state*

$$a(S_0) = \frac{1}{9}, \qquad R_{15}(S_0) = g(15)\frac{4877231181816111466}{252095850530248725}. \tag{10}$$

Proof. Put $M = \binom{\mathbf{1}^T}{B}$, $F = \ker M$, $\lambda = \mathbf{1}/18$, and $V = \mathrm{vol}_{15}(S_0)$. For an orthonormal frame U of F, write the facet area-normal vectors as $-d_i P_F e_i$. Their balance implies $d \in F^\perp = \mathrm{span}(\mathbf{1}, B^T)$. The permutation σ with $b_{\sigma(i)} = -b_i$ preserves the section and gives $d_i = d_{\sigma(i)}$. Since B has rank two, this forces $d = t\mathbf{1}$. Taking the origin at λ, the divergence identity gives $t = 15V$. This permutation symmetry does not imply central symmetry of S_0.

Define the two finite determinant sums

$$D = \sum_{i<j<k} |\det(b_j - b_i, b_k - b_i)| = 101088, \qquad E = \sum_{i<j} |\det(b_i, b_j)| = 11232.$$

Complementary minors of U and $[U, \lambda]$, using $B\mathbf{1} = 0$, give respectively $D/\sqrt{\det(MM^T)}$ and $E/\sqrt{\det(MM^T)}$ for their maximal-minor sums. The facet determinant identities therefore yield

$$R_{15}(S_0) = \frac{15^{15} V D}{\sqrt{\det(MM^T)}}, \qquad a(S_0) = E/D.$$

Let f be the two-dimensional density of $Z = \sum_i b_i X_i$, where the X_i are independent mean-one exponential variables. Coarea and scaling of the fibers $Mx = (s, 0)$ give

$$f(0) = \int_0^\infty e^{-s} \frac{s^{15} V}{\sqrt{\det(MM^T)}}\, ds = \frac{15! V}{\sqrt{\det(MM^T)}}.$$

This is the standard multivariate-spline volume method; see [8]. Thus $R_{15}(S_0) = g(15) D f(0)$.

For completeness, the supplied certificate evaluates the density by a finite rational identity. For every independent pair $Y = (b_i, b_j)$, put $\theta_Y = Y^{-T}\mathbf{1}$ and $\alpha_Y = \prod_{k \notin \{i,j\}} (1 - \theta_Y \cdot b_k)^{-1}$. No three specified columns are affinely collinear. The checker verifies all coefficients of

$$1 = \sum_Y \alpha_Y \prod_{k \notin Y} (1 + w \cdot b_k). \tag{11}$$

Dividing by the common product identifies the Laplace transform of Z with a finite signed sum of transforms of $Y(X_i, X_j)$. Uniqueness of these transforms gives their density identity. The original density is continuous: two independent antipodal pairs already give a bounded continuous planar density, and convolution with the remaining pairs preserves continuity. Taking a generic ray h to zero consequently gives

$$f(0) = \sum_{Y^{-1}h>0} \frac{\alpha_Y}{|\det Y|} = \frac{2438615590908055733}{12741932669200891556400}.$$

The certificate checks three distinct generic rays, with the same exact answer. Substitution proves (10). □

5 A general section seed with stronger growth

The symmetry above permits particularly short coordinates, but retaining the facet weights allows a larger improvement. Let C be the integer 4×23 matrix specified in Appendix B, and put

$$G_0 = \{x \in \mathbb{R}^{23} : x_i \geq 0, \quad \mathbf{1} \cdot x = 1, \quad Cx = 0\}.$$

It has dimension eighteen and twenty-three genuine facets. These claims are checked by exact minors: every four columns are linearly independent, every five are affinely independent, and deleting any column leaves a simplex of columns containing zero strictly in its interior.

Proposition 4. *The seed G_0 satisfies the rational enclosures*

$$\frac{108996733071}{10^{12}} \leq a(G_0) \leq \frac{108996733072}{10^{12}}, \tag{12}$$

$$\frac{194095847511843177}{10^{9}} \leq R_{18}(G_0) \leq \frac{194095847511843178}{10^{9}}. \tag{13}$$

Proof. We give the geometric identities needed to interpret the exact interval certificate. For a rank-r matrix B with $m = n + r + 1$ columns, let $S(B) = \{x \geq 0 : \mathbf{1} \cdot x = 1, Bx = 0\}$, $M = \binom{\mathbf{1}^T}{B}$, and let f be the density of $\sum_i b_i X_i$ for independent mean-one exponentials. Suppose zero lies in the interior of the convex hull after each column deletion, and the density is C^1 at zero. Set

$$\gamma_i = 1 + \frac{b_i \cdot \nabla f(0)}{f(0)},$$

$$D_\gamma = \sum_{|J|=r+1} |\det M_J| \prod_{i \notin J} \gamma_i, \qquad E_\gamma = \sum_{|J|=r} |\det B_J| \prod_{i \notin J} \gamma_i.$$

Here M_J and B_J retain the indicated columns. The weighted form of the section calculation is

$$R_n(S(B)) = g(n) f(0) D_\gamma, \qquad a(S(B)) = E_\gamma / D_\gamma. \tag{14}$$

Indeed, let $V = \mathrm{vol}_n(S(B))$, $F = \ker M$, and write its facet vectors as $-d_i P_F e_i$. Coarea gives $f(0) = n!V/\sqrt{\det(MM^T)}$. If f_i omits the ith exponential summand, integration by parts gives $f_i(0) = f(0) + b_i \cdot \nabla f(0)$. The facet coarea identity and

$$\det(M_{-i} M_{-i}^T) = \det(MM^T) \|P_F e_i\|^2$$

therefore give $d_i = nV\gamma_i$. Applying complementary minors to an orthonormal frame of F and its lift by any λ with $M\lambda = (1, 0)$ proves (14). These are the classical section and spline volume tools [4, 8], with the facet weights retained.

For the specified C, the required C^1 regularity follows directly from Fourier inversion. Any unit vector is orthogonal to at most three columns, so compactness gives a uniform positive lower bound for the fourth smallest absolute scalar product. The characteristic function $\prod_i (1 + \mathrm{i}\, b_i \cdot t)^{-1}$ consequently decays as $O((1 + \|t\|)^{-20})$. It and its first frequency moments are integrable in four dimensions. The same argument applies after deleting a column.

The density and gradient use the same finite Laplace identity as above, now for four-column bases Y. All 8855 linear and 33649 affine bases are nonsingular. To justify the identity without a large coefficient list, put $\theta_Y = Y^{-T}\mathbf{1}$ and $\alpha_Y = \prod_{k \notin Y} (1 - \theta_Y \cdot b_k)^{-1}$. The polynomials $\alpha_Y \prod_{k \notin Y} (1 + w \cdot b_k)$ take the values δ_{YZ} at the nodes $w = -\theta_Z$. Their number is $\binom{m}{r}$, the dimension of polynomials of degree at most $m - r$ in r variables. They form an interpolation basis and sum to one. Laplace uniqueness and C^1 regularity thus give, for a generic ray h,

$$f(0) = \sum_{Y^{-1}h>0} \frac{\alpha_Y}{|\det Y|}, \qquad \nabla f(0) = -\sum_{Y^{-1}h>0} \frac{\alpha_Y \theta_Y}{|\det Y|}.$$

The checker encloses each rational summand outwards at denominator 2^{256}, then encloses the positive facet weights and their products at denominator 2^{128}. It repeats on two generic rays, checks overlap in every component, and substitutes the resulting intervals in (14). Coarsening only outwards gives (12)–(13). No floating-point value is an acceptance premise. □

6 Recursive amplification and exact tail control

Starting from any of the four specified asymptotic seeds, apply the prior scheme of [13, Section 9]:

$$K_{j+1} = (K_j \times K_j) * (K_j \times K_j), \qquad d_j = \dim K_j, \quad R_j = R_{d_j}(K_j), \quad a_j = a(K_j). \tag{15}$$

The inherited identities give

$$d_j = \frac{(3d_0 + 1)4^j - 1}{3}, \qquad a_j = \frac{a_0}{2^j}, \qquad R_{j+1} = R_j^4 D_j, \quad D_j = 2a_j \frac{g(4d_j + 1)}{g(2d_j)^2}. \tag{16}$$

Proposition 5 (Exact seed ranking at fixed dimension)**.** *For any two full-dimensional polytope seeds A, B of the same positive dimension d, the recursion (15) has positive root limits β_A, β_B and*

$$\left(\frac{\beta_A}{\beta_B}\right)^{3d+1} = \frac{a(A)R_d(A)^3}{a(B)R_d(B)^3}. \tag{17}$$

Thus maximizing the limiting rate in the fixed-dimensional family $K(u)$ is exactly equivalent to maximizing the rational objective

$$J(u) = I(u)^3 p(u)^2 s(u).$$

This ranking concerns the specified recursion, not every operation schedule.

Proof. The central-binomial bounds used below imply that D_j lies between two positive constants depending only on d, a_0. The logarithmic recurrence therefore gives a finite root limit for every seed. For equal-dimensional seeds their multiplier ratio is $a(A)/a(B)$ at every level, so exact iteration gives

$$\frac{R_j(A)}{R_j(B)} = \left(\frac{R_d(A)}{R_d(B)}\right)^{4^j} \left(\frac{a(A)}{a(B)}\right)^{(4^j-1)/3}.$$

Take d_jth roots and use $4^j/d_j \to 3/(3d+1)$ to prove (17). Finally $aR^3 = g(d)^3 I^3 p^2 s$ by (5). The factor $g(d)^3$ is constant at fixed dimension. □

Lemma 6. *For the three preceding seeds put $c = 7$, and for G_0 put $c = 8$. For every $j \geq 0$, one has $3 < D_j < c$ and $R_j^{1/d_j} < 3$. The roots R_j^{1/d_j} strictly increase to a limit β with*

$$(3R_j^3)^{1/(3d_j+1)} < \beta < (cR_j^3)^{1/(3d_j+1)}. \tag{18}$$

Proof. We extend the elementary tail argument in [13, Lemma 9.1] to these seed states. For $q \geq 1$,

$$\frac{4^q}{2\sqrt{q}} \leq \binom{2q}{q} \leq \frac{4^q}{\sqrt{3q+1}}.$$

Both follow by induction from equality at $q = 1$. For the lower bound the squared induction comparison has difference $(2q+1)^2 - 4q(q+1) = 1$; for the upper bound it has difference $4(q+1)^2(3q+1) - (2q+1)^2(3q+4) = q$. Set $q = 2d_j$ and note that

$$\frac{g(2q+1)}{g(q)^2} = \frac{4^q}{\binom{2q}{q}} \left(1 + \frac{1}{2q}\right)^{2q}. \tag{19}$$

The last factor is strictly between 2 and e. Thus

$$4a_j\sqrt{6d_j+1} < D_j < 4ea_j\sqrt{2d_j}.$$

The closed dimension formula implies

$$a_j^2(6d_j+1) = a_0^2(6d_0+2-4^{-j}) \geq a_0^2(6d_0+1),$$

and $a_j^2(2d_j) < a_0^2(2d_0+2/3)$. The exact checks

$$16a_0^2(6d_0+1) > 9, \qquad 121a_0^2(2d_0+2/3) < c^2, \qquad cR_0^3 < 3^{3d_0+1} \tag{20}$$

hold for all four seeds; for G_0, use the lower endpoint of (12) in the first comparison and the upper state endpoints in the other two. The elementary bound $e < 11/4$ proves $D_j < c$. Iterating $R_{j+1} < cR_j^4$ gives

$$R_j^3 \leq \frac{(cR_0^3)^{4^j}}{c} < \frac{3^{(3d_0+1)4^j}}{c} = \frac{3^{3d_j+1}}{c} < 3^{3d_j}.$$

Since $D_j > 3 > R_j^{1/d_j}$, the recurrence proves strict root increase. The upper bound by three proves convergence. Iteration of the logarithmic recurrence and the dimension formula yield the absolutely convergent series

$$\log\beta = \frac{\log R_j + \sum_{t\geq 0} 4^{-t-1}\log D_{j+t}}{d_j + 1/3}.$$

Using $3 < D_{j+t} < c$ and $\sum_{t\geq 0} 4^{-t-1} = 1/3$ proves the two strict bounds in (18). □

Proposition 7. *For the twelve-dimensional seed, $2.861 < \beta < 2.8611$. For the eight-dimensional seed, $2.8571 < \beta < 2.8573$. For the fifteen-dimensional seed, $2.8900 < \beta < 2.8902$. For G_0, $2.9280 < \beta < 2.9283$.*

Proof. The finite certificates reconstruct the exact rational state R_5 for the first three seeds and outward state bounds for G_0. The recurrence is increasing in both R and a, so the latter bounds propagate rigorously. The checkers use (19), without importing the producer's factorial formula. They verify by integer comparison

$$\ell^{3d_5+1} < 3R_5^3, \qquad cR_5^3 < h^{3d_5+1},$$

using the lower and upper state bounds, respectively, for G_0, and the following rational endpoints:

Seed dimension	d_5	ℓ	h	c
8	8533	28571/10000	28573/10000	7
12	12629	2861/1000	28611/10000	7
15	15701	289/100	14451/5000	7
18	18773	366/125	29283/10000	8

Together with Lemma 6, these finite inequalities certify the limits. No floating-point logarithm or numerically truncated infinite product is used in acceptance of the endpoints. □

For the asymptotic construction in Theorem 1, take the eighteen-dimensional family. For $n \geq 324$, choose the largest j with $d_j \leq \sqrt{n}$, put $k = \lfloor n/d_j \rfloor$, $r = n - kd_j$, and set

$$A_n = K_j^k \times T_r.$$

Here K_j^k denotes a Cartesian power; the factor T_0 is omitted. For $n < 324$ use $A_n = T_n$. Then $j \to \infty$, $0 \leq r < d_j \leq \sqrt{n}$, and $kd_j/n \to 1$. Product multiplicativity gives

$$\frac{\log R_n(A_n)}{n} = \frac{kd_j}{n}\frac{\log R_j}{d_j} + \frac{\log c_r}{n}.$$

Set $c_0 = 1$. The bound $c_r \leq (r+1)e^r$ makes the last term tend to zero, so the required all-dimensional limit is β. This padding argument is the one used for the prior recursive family in [13, Corollary 9.3].

7 The imported upper bound and separation from dimension nine

The mixed Bellman theorem [10] states, for every $K \in \mathcal{C}$, with $D = n + 1$ and $H = 1/a(K)$,

$$\log Q(K) \leq \alpha\left(D - \frac{H^2}{D}\right) + \delta\left(D - \frac{H^4}{D^3}\right), \qquad \alpha = \frac{87}{2000}, \quad \delta = \frac{11}{2000}. \tag{21}$$

This is an imported theorem, not a new inequality of this note. Its pinned exact certificate was replayed successfully: 19,900 finite state rectangles, 15,568 tail cells, and the analytic large-dimension constants. The complete English argument was also inspected. Exact replay checks its finite proof obligations; it does not replace review of its analytic argument.

Discarding the dependence on H gives the coarse bound $R_n(K) < 1.051(n+1)(2.8548)^n$. Combined with a single block and simplex padding, this gave the earlier sufficient threshold 15500. Retaining that dependence yields a much smaller comparison bound. Write

$$f_n(H) = \log H + \alpha(D - H^2/D) + \delta(D - H^4/D^3).$$

Then $R_n(K) \le g(n) \exp f_n(H)$. For every rational $h > 0$, $f_n''(H) \le -2\alpha/D$ implies

$$\sup_{K \in \mathcal{C}_n} R_n(K) \le g(n) h \exp \left[\alpha(D - h^2/D) + \delta(D - h^4/D^3) + \frac{D}{4\alpha} \left(\frac{1}{h} - \frac{2\alpha h}{D} - \frac{4\delta h^3}{D^3} \right)^2 \right]. \tag{22}$$

Indeed, the quadratic upper Taylor bound at h is maximized by completing the square. Rational centres near its stationary point give tight bounds without solving an algebraic equation in proof acceptance.

Lemma 8. *For every $n \ge 49$,*

$$\sup_{K \in \mathcal{C}_n} R_n(K) < \frac{9}{10} \left(\frac{7137}{2500} \right)^n. \tag{23}$$

Proof. Drop the nonpositive quartic term in $f_n(H)$ and use $xe^{-x} \le e^{-1}$, with $x = 2\alpha H^2/D$. This gives

$$R_n(K)^2 \le \frac{1000}{87}(n+1) g(n)^2 \exp \left(\frac{49}{500}(n+1) - 1 \right).$$

The sequence $B_n = (n+1)g(n)^2 e^{-2n}$ strictly decreases. Its consecutive ratio is $(n+2)(1+1/n)^{2n}/((n+1)e^2)$, which is below one since

$$2n \log(1 + 1/n) < 2 - \frac{1}{n+1}, \qquad \log(1 + 1/(n+1)) < \frac{1}{n+1}.$$

The first inequality follows by integrating $t/(1+t) \ge t/(1+x)$ on $0 \le t \le x$, with $x = 1/n$. Consequently, for $n \ge 49$,

$$R_n(K)^2 \le \frac{1000}{87} B_{49} e^{-451/500} e^{(1049/500)n}.$$

Rational Taylor bounds certify

$$\frac{1000}{87} 50 g(49)^2 e^{-98-451/500} < \frac{81}{100}, \qquad e^{1049/1000} < \frac{7137}{2500}.$$

Taking positive square roots proves the claim. In particular $\Gamma_{\mathcal{C}} < 2.855$, as used in Theorem 1. □

Two additional rational members of the inherited seed family suffice for the finite witnesses. The ten-dimensional seed is

$$S_{10} = K(16/1000, 51/1000, 91/1000, 142/1000, 217/1000).$$

The nine-dimensional seed S_9 is a pyramid over $K(1/100, 2/100, 3/100, 5/100)$ in dimension eight; equivalently, append one zero coefficient to its paired tilt vector. It has 26 vertices and 11 facets, whereas S_{10} has 36 vertices and 12 facets. The calculus in (H, Q) extends to a point with the formal values $H = Q = 1$:

$$(H, Q)(A * B) = (H_A + H_B, Q_A Q_B), \tag{24}$$

$$H(A \times B) = \frac{r+s}{r/H_A + s/H_B}, \qquad Q(A \times B) = Q_A Q_B \frac{g(r)g(s)}{g(r+s)} \frac{sH_A + rH_B}{r+s}. \tag{25}$$

The point case in (24) is the ordinary pyramid formula; products with a point can be omitted. These are rewritings of the prior operation rules, not new geometric identities.

Proposition 9. *The accompanying 420-node rational operation certificate specifies polytopes W_n for $9 \le n \le 245$, built from points, S_9, S_{10}, and K_0 by products and joins. Let $L = K_2$, of dimension 197. Define*

$$E_n = \begin{cases} W_n, & 9 \le n \le 48, \\ L^k \times W_s, & n \ge 49, \quad s = 49 + ((n-49) \bmod 197), \quad k = (n-s)/197. \end{cases}$$

Then $R_n(E_n) > \sup_{K \in \mathcal{C}_n} R_n(K)$ for every $n \ge 9$. The factor L^0 is omitted.

Proof. For $n = 9$, the prior exact Pareto classification gives $\sup_{K \in \mathcal{C}_9} R_9(K) = c_9$ [11, Theorem 1]. Our checker replays this small instance: starting with $(H, Q) = (1, 1)$, it enumerates every product and join of smaller-dimensional Pareto states, then discards coordinatewise dominated states. Both outputs in (24)–(25) increase in both input coordinates at fixed dimensions, so induction proves completeness. The replay checks 261 candidates, retaining 50 states across dimensions zero through nine. The independent geometry check gives

$$\frac{R_9(S_9)}{c_9} = \frac{20811283583}{20797677600} > 1.$$

Set $W_9 = S_9$. The geometric family and the dimension-nine simplex counterexample phenomenon are already prior work [4].

For $10 \le n \le 48$, the certificate specifies W_n and a rational centre h_n. Reconstructing every operation state and applying rational upper Taylor intervals for the exponential in (22) gives the strict comparison in each of these 39 dimensions. For each of the 197 integers $49 \le s \le 245$, it instead certifies

$$R_s(W_s) > \frac{1001}{1000}\frac{9}{10}\left(\frac{7137}{2500}\right)^s.$$

The dimension ranges and certified multiplicative margins are:

Dimensions	Class upper bound	Witness / upper bound
9	c_9, exact Pareto replay	$> 4001/4000$
10–48	Rational tangent bounds	$> 4001/4000$
49–245	$(9/10)(7137/2500)^n$	$> 1001/1000$

These are verified lower margins, not approximations to sharp extrema. For example, the recipes choose $W_{10} = S_{10}$, W_{18} as six successive pyramids over K_0, and $W_{49} = K_1$. The finite recipes are explicit construction witnesses; completeness of the search that discovered them is unnecessary.

Finally the exact recurrence gives

$$R_{197}(L) > \frac{5}{4}\left(\frac{7137}{2500}\right)^{197}.$$

Product multiplicativity and Lemma 8 now prove the claim for every $n = s + 197k \ge 49$. This covers every residue and every larger dimension, with no finite truncation of the conclusion. The block retains its explicit 224-facet description and has 46,118,408 vertices, counted by the recursion rather than enumerated. □

To obtain one sequence satisfying both assertions of Theorem 1, set $E_n = T_n$ for $n < 9$ and choose P_n to be whichever of A_n, E_n has larger R_n, breaking ties in favour of A_n. This is an exact rational comparison. Since $R_n(E_n)^{1/n} \to R_{197}(L)^{1/197} < 2.8611 < \beta$ by strict root increase for the twelve-dimensional family and its certified enclosure, whereas $R_n(A_n)^{1/n} \to \beta$, the selected sequence has limit β and strictly exceeds the entire class in every dimension $n \ge 9$.

8 Exact verification

The principal seed is checked in `research/section-seeds/`. The checker verifies every linear and affine basis, proves all facet hypotheses by exact barycentric signs, and reconstructs the density, gradient, facet weights, state enclosures and amplified endpoints with outward integer arithmetic. Six additional adversarial and regression tests run in normal and optimized Python.

The simpler symmetric seed is checked in `research/polygon-seeds/`. The standard-library checker reconstructs the eighteen integer columns, verifies all facet hypotheses, checks 153 coefficients of the cleared Laplace identity from 144 independent pairs, and computes the density on three generic rays. It then checks the determinant sums, exact seed state, tail hypotheses, and the rational growth endpoints. Five adversarial and regression tests run in normal and optimized Python. This certificate verifies the finite identities in Proposition 3; their geometric interpretation is proved there rather than inferred from numerical search.

The accompanying standard-library Python package contains the four exact two-upper-facet seeds, their vertex lists and oriented triangulations, the full rational recurrence states, and separate geometric and growth checkers. The producers use pulling triangulations; the standard geometric framework is described in [3]. The checkers validate the resulting oriented chains independently of their construction. For the twelve-dimensional seed, the geometric checker solves all 91 active systems, verifies 924 positively oriented simplices, cancels all interior boundary faces, and verifies multiplicity one at a generic rational interior point. The analogous counts for the eight-dimensional seed are 45 active systems and 70 simplices. All uncancelled faces lie on the body's boundary. Crossing an interior codimension-one face therefore leaves the positive-chain multiplicity unchanged, and a generic interior path avoids faces of codimension two. Multiplicity is constant in the interior; its checked value one proves coverage almost everywhere with no interior overlap.

The checker obtains volumes from simplex determinants and facet vectors from boundary cofactors, then sums every horizontal and lifted maximal minor. Thus it checks both R_0 and a_0 without assuming Proposition 2. The recurrence checker imports neither the geometric producer nor its formula module and uses central binomial coefficients instead of factorial ratios. It checks (20), the rational limit endpoints and the original dimension-197 block comparison. A separate finite-separation checker reconstructs all 420 witness nodes, replays the exact dimension-nine class frontier, checks the 39 rational tangent bounds and all 197 residue witnesses, and verifies the finite constants in the analytic all-dimensional extension. The additional nine- and ten-dimensional two-upper-facet seeds have 55 and 66 active systems and 70 and 252 oriented simplices, respectively. Explicit exceptions keep the mathematical checks active under optimized Python. The 39 finite comparisons are also cross-checked using rational atanh intervals for logarithms, without reusing the exponential-series code. The dimension-197 block also has an explicit inequality certificate. A separate checker compares its 224 inequalities with a flattened description using sixteen seed copies, four inner join coordinates, and one outer join coordinate. It verifies a strict interior point. This checks equivalence to the recursive expression; it does not enumerate the block's vertices or independently recompute its volume.

The runner in `reproduction/` verifies the earlier seeds, finite comparisons, and sixteen unit tests. Invoke it with `python verify_all.py`. The runner writes regenerated certificates and reports to temporary files and leaves the delivered files unchanged. Empty or incomplete delivery manifests and mathematically false comparison constants are rejected. The command `python replay_prior.py` fetches byte-pinned external upper-bound inputs and replays their checker. This optional step needs network access on first use. The delivery includes its successful replay report and the input manifest. Formula and geometric data, unlike PDF build metadata, regenerate byte-identically in the tested environments.

9 Discussion

The separation in Theorem 1 identifies a limitation of the point-generated class rather than a failure of products and joins themselves. The same inherited operations amplify the new seeds beyond the upper rate available to every finite expression in $\mathcal{C}$. Thus changing the initial geometric input can matter even when the subsequent construction rules remain fixed. The result compares complete construction classes; exceeding one previously known recursive sequence alone would not establish this conclusion.

The eight-dimensional example illustrates why immediate seed performance can be misleading. Its ratio R_8/c_8 is below one, yet its recursively amplified rate exceeds the ceiling for $\mathcal{C}$. The lifted invariant a records information used by joins that the initial value of R alone does not capture. Proposition 5 makes this observation quantitative: at a fixed dimension and for the specified recursion, the correct seed objective is aR^3, or equivalently I^3p^2s in the stated tilt family. This supplies an exact target for further seed search. It does not justify comparing different dimensions by that objective or claiming that the balanced recursion is the best possible schedule.

The general section seed raises the certified amplified lower rate to 2.928, compared with 2.890 for the polygon section and 2.861 for the two-upper-facet seed, using the same recursion. Its twenty-three facets are five more than its dimension. Retaining the nonuniform facet weights allows a larger improvement, at the cost of less compact seed coordinates. The polygon example offers a simpler symmetric comparison. Its eighteen facets are only three more than its dimension, but their planar section constraints allow geometric inputs outside the preceding two-upper-facet family. This shows a concrete benefit from broadening the seed search. These certificates establish particular witnesses and their rates. Optimality of either family and necessity of the additional facets remain open.

The asymptotic advantage is small at the level of a single root, but it produces an exponential ratio of volumes as dimension grows. The finite construction now separates in every dimension at least nine. The reduction from the earlier sufficient threshold 15500 comes from retaining the invariant dependence in the class envelope and replacing a uniform simplex padding loss by finitely many checked product–join witnesses. It does not improve the imported asymptotic class ceiling. The separate section constructions strengthen the certified value of β. Different finite and asymptotic objectives favour different seeds. The threshold nine is not proved minimal: the present result makes no new assertion about unrestricted maximizers in dimensions four through eight. Nor does the lower rate approach a known unrestricted optimum: the classical upper root $\pi e/2$ leaves a substantial gap.

Explicitness also needs a computational qualification. The block L has a compact description by 224 facet inequalities but more than 46 million vertices. Its certified ratio is obtained from the checked seed geometry and the inherited operation identities, rather than by enumerating its vertices or constructing its projection body directly. The package therefore makes these particular witnesses and their comparison independently checkable; it does not provide an efficient algorithm for optimizing projection-body volume over arbitrary polytopes.

The two-upper-facet family and section method are prior work [4], and the spline volume method is classical [8], as are the lifted invariant, operation rules, recursion, and tail method [13]. The whole-class upper theorem [10] is an essential external dependency. Its exact finite obligations were replayed, but this does not formalize its analytic argument or constitute external peer review. The contribution relative to the inspected sources is the specified rational seeds, their certified amplified rates, and the resulting class separation and effective comparison. The supplementary literature audit does not establish bibliographic priority.

Further progress could take several forms. A structural bound for the fixed-dimensional objective

aR^3 over a specified section family would distinguish a good rational seed from an optimum. Enlarging that family, changing seed dimension, or allowing unbalanced schedules may improve the lower rate. Sharper class envelopes may reduce the finite certificate or improve separation margins. Moving below dimension nine would require a new unrestricted counterexample, not merely a smaller padding constant. These optimization problems are not resolved by the present certificates. External mathematical review and proof-assistant formalization would strengthen confidence in both the new argument and its imported dependencies.

10 Conclusion

An explicit rational eighteen-dimensional simplex-section seed, amplified by the prior product–join recursion, gives a projection-volume root limit strictly between 2.9280 and 2.9283. Combined with the attributed upper theorem for the point-generated class, this proves a strict exponential separation from that entire class. A finite rational operation certificate and a repeating 197-dimensional block give a comparison in every dimension $n \geq 9$. One explicit sequence satisfies this finite separation while retaining the stronger root limit.

The examples show that the initial seed and its lifted invariant are material to recursive growth. Exact geometry, recurrence certificates, and a finite comparison make the claims reproducible without treating numerical search as proof. The unrestricted extremal rate, optimal seed and schedule, minimal separation dimension, and global bibliographic priority remain open.

Generative AI disclosure

OpenAI Codex agents contributed substantially to source discovery, literature synthesis, and mathematical exposition.

Data and code availability

The manuscript source, PDF, exact seed coordinates, triangulations and section certificates, dimension-197 facet inequalities, finite operation recipes, recurrence certificates, verification programs, source audit, and reproduction instructions are publicly available in the accompanying GitHub repository [5]. The versioned release `v1.3.0` identifies the edition accompanying this manuscript. Original Python code and generated certificate data are provided under the MIT license; manuscript rights are stated separately.

From the release's `reproduction/` directory, run `python -B verify_all.py --regenerate` with Python 3.11 or newer. All checks of the new constructions use only the standard library and run without network access. The optional flag `--with-prior` fetches and replays the byte-pinned external upper-theorem package; its first use requires network access. The delivery manifest records SHA-256 hashes, and verification leaves delivered files unchanged. The release includes the successful external replay record. External inputs remain attributed and are fetched from their pinned upstream locations rather than redistributed as part of this package. From the repository root, also run `python -B research/polygon-seeds/check_polygon_seed.py --check-report` for the symmetric polygon seed. For the principal seed and its stronger root enclosure, run `python -B research/section-seeds/check_section_seed.py --check-report`.

A A simplex truncation identity

The following is a standard simplex section formula; see Lasserre [6, Theorem 2.2] and the Genocchi–Hermite representation of divided differences [2, Section 9]. We use the following normalization. Let $t_0, \dots, t_n$ be distinct real numbers, and let $\Delta_n = \{y_i \geq 0 : \sum_{i=1}^n y_i \leq 1\}$, with $y_0 = 1 - \sum_{i=1}^n y_i$. Then

$$\frac{\mathrm{vol}_n\{y \in \Delta_n : \sum_{i=0}^n t_i y_i \geq 0\}}{\mathrm{vol}_n(\Delta_n)} = \sum_{t_i > 0} \frac{t_i^n}{\prod_{j \neq i}(t_i - t_j)}. \tag{26}$$

Here a zero node contributes zero.

B Integer data for the eighteen-dimensional seed

Write $C = (C_1, C_2)$, where the columns are specified by the following two transposed blocks, in the indicated order:

$$C_1^T = \begin{pmatrix} 670 & 2056 & -86 & 672 \\ 360 & 1447 & 393 & -873 \\ -930 & 514 & -291 & 1940 \\ -1431 & 154 & 655 & -613 \\ 1540 & 969 & -1320 & -1634 \\ -88 & -1771 & -1627 & -603 \\ -54 & -72 & -562 & -2830 \\ 2577 & 679 & -5 & 216 \\ -1190 & 1180 & -1095 & -677 \\ -64 & -1434 & 1214 & 372 \\ -1774 & -1637 & 128 & -1102 \\ 2009 & -121 & 669 & -1194 \end{pmatrix}, \qquad C_2^T = \begin{pmatrix} 1637 & -1657 & -39 & -33 \\ -2480 & -740 & -881 & -186 \\ 538 & 579 & -1385 & 823 \\ 1046 & 1303 & 1133 & 1626 \\ -570 & 59 & 1155 & 1980 \\ -868 & -739 & -2097 & -2323 \\ 354 & -1960 & -445 & -2363 \\ -1386 & 1039 & 482 & 1000 \\ 554 & 276 & 1576 & 387 \\ 1284 & -5 & 511 & 2049 \\ -1183 & -1571 & 106 & 1308 \end{pmatrix}.$$

The release stores these same integers in `research/section-seeds/seed.json`. The checker verifies all basis determinants, 3113 strictly interior simplex bases, and every coordinate facet. The generic rays $(17, 13, 7, 3)$ and $(19, 11, 5, 2)$ give overlapping outward enclosures for the density, gradient and complete geometric state. The matrix specifies a witness; no seed or combinatorial optimality is claimed.

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